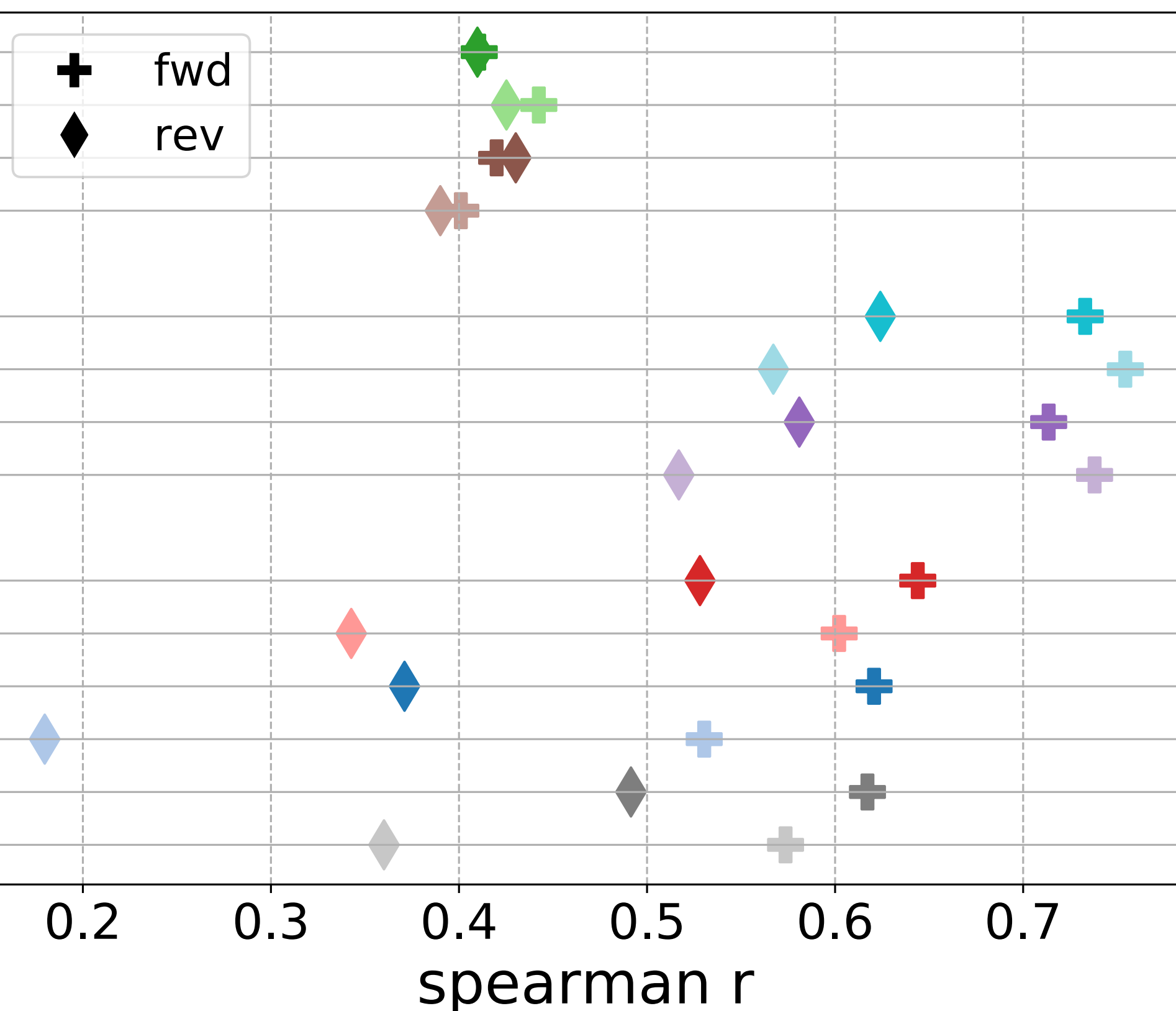


A HERMES-fixed Untr. 0.50 + Megascale
 HERMES-fixed Untr. 0.00 + Megascale
 HERMES-fixed Untr. 0.50 + cDNA117k
 HERMES-fixed Untr. 0.00 + cDNA117k

HERMES-fixed 0.50 + Megascale
 HERMES-fixed 0.00 + Megascale
 HERMES-fixed 0.50 + cDNA117k
 HERMES-fixed 0.00 + cDNA117k

HERMES-amortized 0.50
 HERMES-amortized 0.00
 HERMES-fixed 0.50
 HERMES-fixed 0.00
 ProteinMPNN 0.30
 ProteinMPNN 0.02

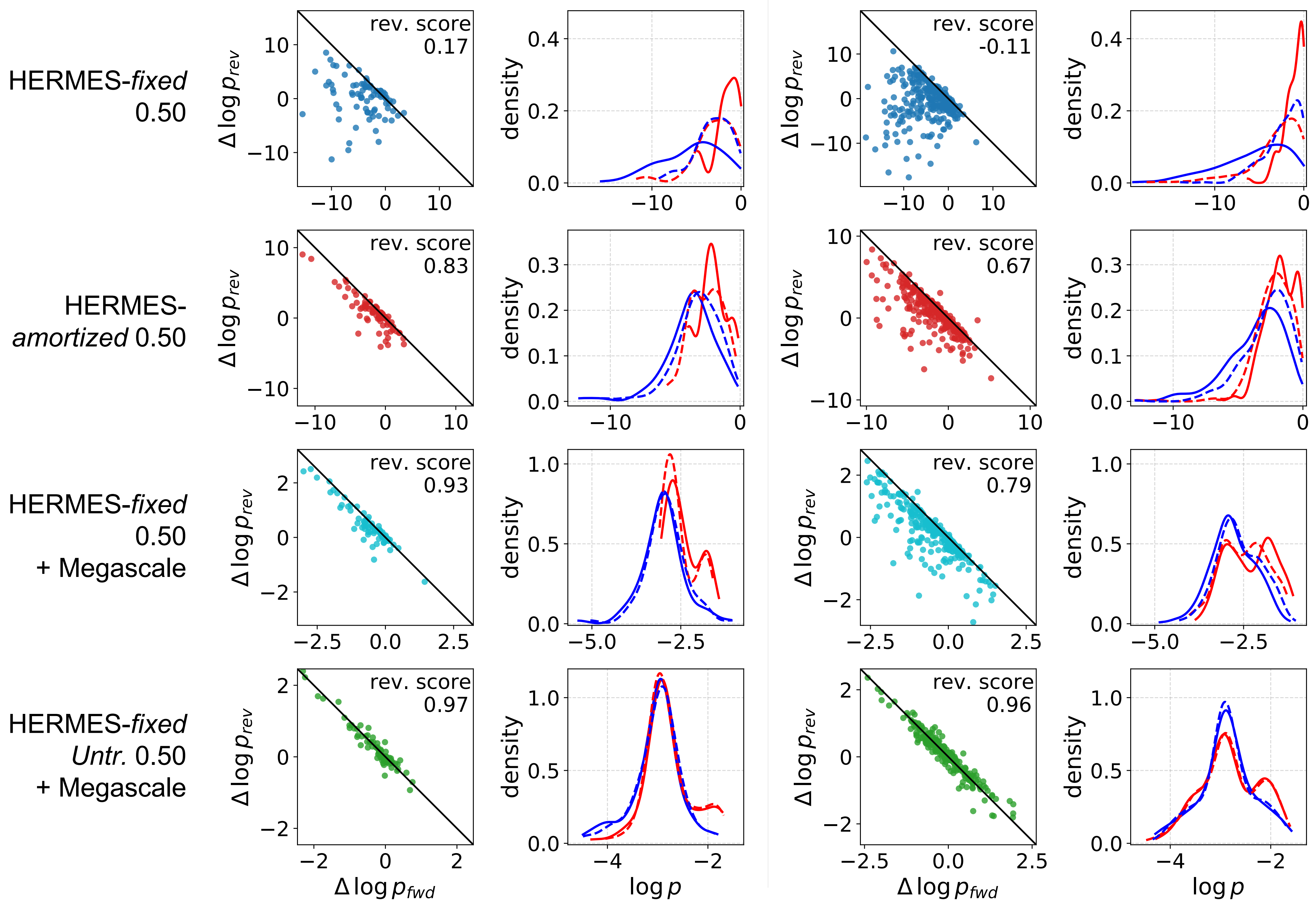


B

max. sequence similarity
to pre-training proteins

< 70%

> 70%



— $\log p(wt | X_{wt})$ - - - $\log p(wt | X_{mt})$
 — $\log p(mt | X_{wt})$ - - - $\log p(mt | X_{mt})$

$\Delta \log p_{fwd} = \log p(mt | X_{wt}) - \log p(wt | X_{wt})$
 $\Delta \log p_{rev} = \log p(wt | X_{mt}) - \log p(mt | X_{mt})$